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room[position] = "Clean"

elif room[position] == "Clean":
    if position == "A":
        print("\nRoom A is Clean")
        print("Action: MOVE RIGHT")
        position = "B"
    else:
        print("\nRoom B is Clean")
        print("Action: MOVE LEFT")
        position = "A"

print("Current State:", room)
print("Vacuum Position:", position)

print("\nBoth rooms are Clean!")
print("Goal State Reached.")

```

vacuum_cleaner()

Initial State:

{'A': 'Dirty', 'B': 'Dirty'}

Vacuum Position: A

Room A is Dirty

Action: SUCK

Current State: {'A': 'Clean', 'B': 'Dirty'}

Vacuum Position: A

Room A is Clean

Action: MOVE RIGHT

Current State: {'A': 'Clean', 'B': 'Dirty'}

Vacuum Position: B

Room B is Dirty

Action: SUCK

Current State: {'A': 'Clean', 'B': 'Clean'}

Vacuum Position: B

Both rooms are Clean!

Goal State Reached.